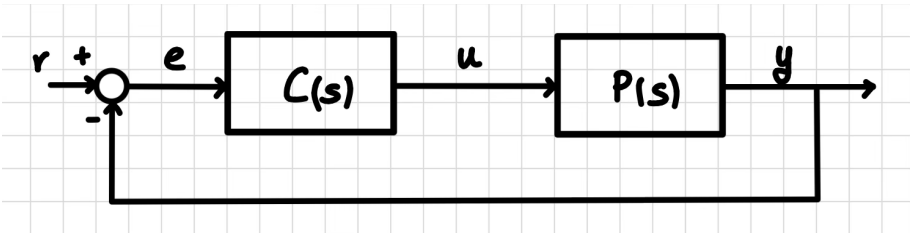
1 Closed-loop system

Consider the closed-loop system



where

- $C(s)$ is the controller
- $P(s)$ is the plant
- $r(s)$ is the reference signal
- $u(s)$ is the plant input
- $y(s)$ is the plant output
- $e(s) = y(s) - r(s)$ is the tracking error.

Definition 1. *The **open-loop transfer function** is*

$$L(s) \;=\; C(s)P(s)$$

Definition 2. *The **sensitivity function S** is the transfer function from the reference to the tracking error.*

$$e(s) \;=\; S\,r(s)$$

We have

$$\begin{aligned} e(s) &= r(s) - y(s) \\ &= r(s) - C(s)\,P(s)e(s) \\ (1 + C(s)P(s))e(s) &= r(s) \\ e(s) &= \frac{1}{1 + C(s)P(s)} r(s) \\ &= \frac{1}{1 + L(s)} r(s) \end{aligned}$$

Therefore, the system's sensitivity function S is

$$S \;=\; \frac{1}{1 + L(s)}$$

Definition 3. *The **complementary sensitivity function** $\boldsymbol{T}$ is the transfer function from the reference to the output.*

$$y(s) \;=\; Tu(s)$$

We have

$$\begin{aligned} e(s) &= r(s) - y(s) \\ y(s) &= r(s) - e(s) \\ &= r(s) - \frac{1}{1+L(s)} r(s) \\ &= \frac{\cancel{1} + L(s) \cancel{-1}}{1+L(s)} r(s) \\ &= \frac{L(s)}{1+L(s)} r(s) \end{aligned}$$

The system's complementary sensitivity function T is therefore

$$T \;=\; \frac{L(s)}{1+L(s)}$$

2 Robust stability

Definition 4. *Given a **family of plants $\mathcal{P}$**, controller C is robust with respect to a property if that property holds for every plant in $\mathcal{P}$.*

The larger the family of plants for which the property holds, the more robust the controller.

Theorem 5. ***(Nyquist criterion)***

Given the open-loop transfer function $L(s) = C(s)\,P(s)$, and assuming there are no inadmissible pole-zero cancellations, define

- *P as the number of poles of $L(s)$ with positive real part*
- *N as the number of counterclockwise encirclements of the point -1 by the Nyquist plot of $L(s)$. If the plot of $L(s)$ passes through -1, N is undefined.*

The closed-loop system is asymptotically stable $\Leftrightarrow$ N is well defined and $N = P$.

Theorem 6.

Consider the family of plants

$$\mathcal{P} \;=\; \{\bar{P}(s) \quad / \quad \bar{P}(s) = P(s)(1 + \Delta(s) \cdot W_2(s))\}$$

where

- *$P(s)$ is the nominal plant, which is asymptotically stabilised by controller $C(s)$*
- *$\Delta(s)$ is a stable, uncertain transfer function satisfying*

$$\|\Delta\|_\infty \;\leq\; 1$$

- *$W_2(s)$ is a weighting function that specifies the maximum size of the uncertainty.*

Define

$$L(s) \;=\; C(s)\,P(s)$$

and define the nominal complementary sensitivity function

$$T(s) \;=\; \frac{L(s)}{1 + L(s)}$$

The family $\mathcal{P}$ is robustly stable $\Leftrightarrow$

$$\|W_2\,T\|_\infty \;<\; 1$$

where T is the complementary sensitivity function associated with plant P.

Proof.

For the perturbed system, the open-loop transfer function is

$$\begin{aligned} \bar{L}(s) \;&=\; C(s)\bar{P}(s) \\ &=\; C(s)P(s)(1 + \Delta(s)\,W_2(s)) \\ &=\; L(s)(1 + \Delta(s)W_2(s)) \end{aligned}$$

Let $\lambda \in [0,1]$. Define

$$\bar{L}(s) \;=\; L(s)(1+\lambda\Delta(s)W_2(s))$$

Then

- if $\lambda = 0$,

$$\bar{L}(s) \;=\; L(s)$$

- if $\lambda = 1$,

$$\bar{L}(s) \;=\; L(s)(1+\Delta W_2)$$

By the Nyquist criterion, asymptotic stability requires

$$L(s)(1+\lambda\Delta(s)W_2(s)) \;\neq\; -1 \qquad \forall\lambda\in[0,1]$$

If the Nyquist plot of $\bar{L}(s)$ passes through the point -1, the number of encirclements changes and the closed-loop system loses asymptotic stability.

Since $(1+L) \neq 0$ by assumption, factoring out $1+L(s)$ gives

$$\begin{aligned}
1+L(s)(1+\lambda\Delta(s)W_2(s)) &\neq 0 \qquad \forall\lambda\in[0,1]\\
1+L(s)+\lambda\,\Delta(s)W_2(s)\,L(s) &\neq 0 \qquad \forall\lambda\in[0,1]\\
(1+L(s))\left(1+\frac{L(s)}{1+L(s)}\lambda\Delta(s)W_2(s)\right) &\neq 0 \qquad \forall\lambda\in[0,1]\\
(1+L(s))(1+T(s)\lambda\Delta(s)W_2(s)) &\neq 0 \qquad \forall\lambda\in[0,1]\\
1+T(s)\lambda\Delta(s)W_2(s) &\neq 0 \qquad \forall\lambda\in[0,1]
\end{aligned}$$

$\Rightarrow$

First, assume the perturbed system is asymptotically stable and prove the implication $\Rightarrow$.

$$\|TW_2\|_\infty \;<\; 1$$

If the perturbed system is asymptotically stable, then

$$1+T(s)\lambda\Delta(s)W_2(s) \;\neq\; 0 \qquad \forall\lambda\in[0,1]$$

For a given $\lambda = 1$, suppose for contradiction that

$$\|TW_2\|_\infty \;\geq\; 1$$

Then $\exists\bar{\omega}$ /.

$$\|T(j\bar{\omega})W_2(j\bar{\omega})\| \;=\; 1$$

Set

$$\Delta(j\bar{\omega}) \;=\; -T(j\bar{\omega})W_2(j\bar{\omega})$$

to obtain

$$\begin{aligned} 1+T(j\bar{\omega})\Delta(j\bar{\omega})W_2(j\bar{\omega}) &= 1-(T(j\bar{\omega})W_2(j\bar{\omega}))^2 \\ &= 1-1 \\ &= 0 \end{aligned}$$

This is a contradiction, which proves the implication.

$\Leftarrow$

We now show that if $\|TW_2\|<1$, then the perturbed system is asymptotically stable.

By the triangle inequality,

$$|a+b| \;\leq\; |a|+|b|$$

it follows that

$$\begin{aligned} 1 &= \|1+T\Delta W_2 - T\Delta W_2\| \\ &\leq \|1+T\,\Delta\,W_2\| + \|-T\,\Delta\,W_2\| \\ &= \|1+T\,\Delta\,W_2\| + \|T\,\Delta\,W_2\| \end{aligned}$$

using $\|\Delta\|_\infty \leq 1$ and $\|TW_2\|<1$.

$$\begin{aligned} 1 &\leq \|1+T\,\Delta\,W_2\| + \|T\,\Delta\,W_2\| \\ &\leq \|1+T\,\Delta\,W_2\| + \|TW_2\| \\ &< \|1+T\,\Delta\,W_2\| + 1 \\ \cancel{1} &< \|1+T\,\Delta\,W_2\| + \cancel{1} \\ 0 &< \|1+T\,\Delta\,W_2\| \end{aligned}$$

The result follows from the Nyquist criterion. □

Remark 7. If

$$\|\Delta\|_\infty \;\leq\; \beta$$

and we set

$$\tilde{\Delta} \;=\; \frac{\Delta}{\beta}$$

then

$$\begin{aligned} \bar{P} &= (1+W_2\Delta)P \\ &= (1+W_2\tilde{\Delta}\beta)P \end{aligned}$$

Robust stability of the family therefore requires

$$\|\beta W_2\,T\|_\infty \;<\; 1$$

3 Robust performance

Consider the perturbed plant $\bar{P}$ with sensitivity function

$$\begin{aligned}
\bar{S} &= \frac{1}{1+\bar{L}} \\
&= \frac{1}{1+L(1+\Delta W_2)} \\
&= \frac{1}{1+L+L\Delta W_2} \\
&= \frac{1}{(1+L)\Big(1+\frac{L}{1+L}\Delta W_2\Big)} \\
&= \frac{1}{(1+L)(1+T\,\Delta\,W_2)} \\
&= \frac{S}{(1+T\,\Delta\,W_2)}
\end{aligned}$$

To ensure that

$$\begin{aligned}
\|\bar{S}\| &< \frac{1}{\|W_1\|} \\
\|\bar{S}W_1\| &< 1
\end{aligned}$$

we require

$$\left\| \frac{S}{(1+T\,\Delta\,W)}\,W_1 \right\| < 1$$

Since

$$\begin{aligned}
1 &= \|1+T\,\Delta\,W_2 - T\,\Delta\,W_2\| \\
&\leq \|1+T\,\Delta\,W_2\| + \|-T\,\Delta\,W_2\| \\
&= \|1+T\,\Delta\,W_2\| + \|T\,\Delta\,W_2\| \\
1-\|T\,\Delta\,W_2\| &\leq \|1+T\,\Delta\,W_2\| \\
\|1+T\,\Delta\,W_2\| &\geq 1-\|T\,\Delta\,W_2\|
\end{aligned}$$

if $\|\Delta\|_\infty \leq \beta \quad \Rightarrow \quad -\|\Delta\|_\infty \geq -\beta$, then

$$\begin{aligned}
\|1+T\,\Delta\,W_2\| &\geq 1-\|T\,\Delta\,W_2\| \\
&\geq 1-\beta\|TW_2\| \\
&= 1-\|\beta TW_2\| \\
\frac{1}{1-\|\beta TW_2\|} &\geq \frac{1}{\|1+T\,\Delta\,W''\|} \\
\frac{1}{\|1+T\,\Delta\,W_2\|} &\leq \frac{1}{1-\|\beta TW_2\|}
\end{aligned}$$

It follows that

$$\left\| \frac{S}{(1+T\,\Delta\,W_2)}\,W_1 \right\| \leq \frac{\|SW_1\|}{1-\|\beta TW_2\|}$$

A sufficient condition for

$$\left\| \frac{S}{(1+T\,\Delta\,W_2)}\,W_1 \right\| < 1$$

is

$$\begin{aligned} \frac{\|SW_1\|}{1-\|\beta TW_2\|} &< 1 \\ \|SW_1\| &< 1-\|\beta TW_2\| \\ \|SW_1\|+\|\beta TW_2\| &< 1 \end{aligned}$$

In summary,

Theorem 8. *given the family of plants*

$$\mathcal{P} \;=\; \{\bar{P}(s) \quad / \quad \bar{P}(s)=P(s)(1+\Delta(s)\cdot W(s))\}$$

where

- $P(s)$ *is the nominal plant, which is asymptotically stabilised by controller* $C(s)$
- $\Delta(s)$ *is a stable, uncertain transfer function satisfying*

$$\|\Delta\|_\infty \;\leq\; \beta$$

- $W_2(s)$ *is a weighting function that specifies the maximum size of the uncertainty,*

define

$$L(s) \;=\; C(s)\,P(s)$$

the nominal sensitivity function

$$S(s) \;=\; \frac{1}{1+L(s)}$$

and the nominal complementary sensitivity function

$$T(s) \;=\; \frac{L(s)}{1+L(s)}$$

If

$$\|SW_1\|+\|\beta TW_2\| \;<\; 1$$

then the family of systems is robustly stable and satisfies

$$\|\bar{S}\|_\infty \;<\; \frac{1}{\|W_1\|_\infty}$$

Definition 9. *The robust performance level* α *is*

$$\alpha \;=\; \max_\omega \|\bar{S}W_1\|$$

4 Assignment 6

Consider the family of plants

$$\tilde{P} \;=\; P(1+\Delta W_2) \qquad \|\Delta\|_\infty \leq 2$$

with

$$P(s)=\frac{1}{s-1} \qquad W_2(s)=\frac{2}{s+10} \qquad C(s)=k \qquad W_1(s)=\frac{1}{s+1}$$

Determine the range of controller gains for which the family of plants is robustly stable.

Find the value of k that guarantees robust stability and minimises the robust performance level α.

5 Robust stability

5.1 Nominal stability

First, determine the values of k that guarantee nominal stability.

The open-loop transfer function is

$$\begin{aligned} L(s) &= C(s)P(s) \\ &= \frac{k}{s-1} \end{aligned}$$

It follows that the sensitivity function is

$$\begin{aligned} S(s) &= \frac{1}{1+L(s)} \\ &= \frac{1}{1+\frac{k}{s-1}} \\ &= \frac{s-1}{s+k-1} \end{aligned}$$

and the complementary sensitivity function is

$$\begin{aligned} T(s) &= \frac{L(s)}{1+L(s)} \\ &= L(s)\cdot S(s) \\ &= \frac{k}{s\neq 1}\cdot\frac{s\neq 1}{s+k-1} \\ &= \frac{k}{s+k-1} \end{aligned}$$

The pole of the complementary sensitivity function is

$$\begin{aligned} s+k-1 &= 0 \\ s &= 1-k \end{aligned}$$

To ensure $\mathrm{Re}(s)<0$, we require

$$\begin{aligned} 1-k &< 0 \\ k &> 1 \end{aligned}$$

5.2 Robust stability

Robust stability requires

$$\|\beta W_2(j\omega)T(j\omega)\| < 1 \qquad \forall\omega$$

where

$$\begin{aligned}\|\beta W_2(j\omega)T(j\omega)\| &= \left\| 2 \cdot \frac{2}{s+10} \cdot \frac{k}{s+k-1} \right\|_{s=j\omega} \\ &= \left\| \frac{4k}{(s+10)(s+k-1)} \right\|_{s=j\omega} \\ &= \left\| \frac{4k}{(j\omega+10)(j\omega+k-1)} \right\| \\ &= \frac{4k}{\sqrt{(\omega^2+100)(\omega^2+(k-1)^2)}}\end{aligned}$$

Since the denominator increases with frequency, $\|\beta W_2(j\omega)T(j\omega)\|$ decreases.

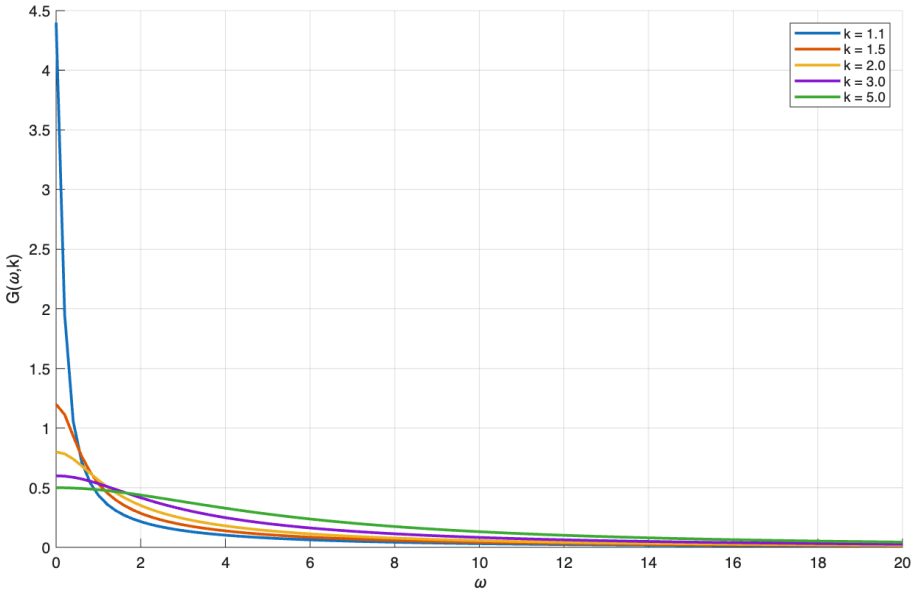


Figure 1. Plot of $\|\beta W_2(j\omega)T(j\omega)\|$ for selected fixed values of k

The maximum of $\|\beta W_2(j\omega)T(j\omega)\|$ is therefore attained at $\bar{\omega}=0$.

$$\begin{aligned}\|\beta W_2(0)T(0)\| &= \frac{4k}{\sqrt{(0+100)(0+(k-1)^2)}} \\ &= \frac{4k}{\sqrt{100(k-1)^2}} \\ &= \frac{4k}{10|k-1|} \\ &= \frac{2k}{5|k-1|}\end{aligned}$$

For $k>1 \quad |k-1|=k-1$, we obtain

$$\|\beta W_2(0)T(0)\| = \frac{2k}{5(k-1)}$$

To ensure

$$\|\beta W_2(0)T(0)\| < 1$$

we therefore require

$$\begin{aligned} \frac{2k}{5(k-1)} &< 1 \\ 2k &< 5(k-1) \\ 2k &< 5k-5 \\ -3k &< -5 \\ k &> \frac{5}{3} \end{aligned}$$

The family of plants is therefore robustly stable if

$$k > \frac{5}{3}$$

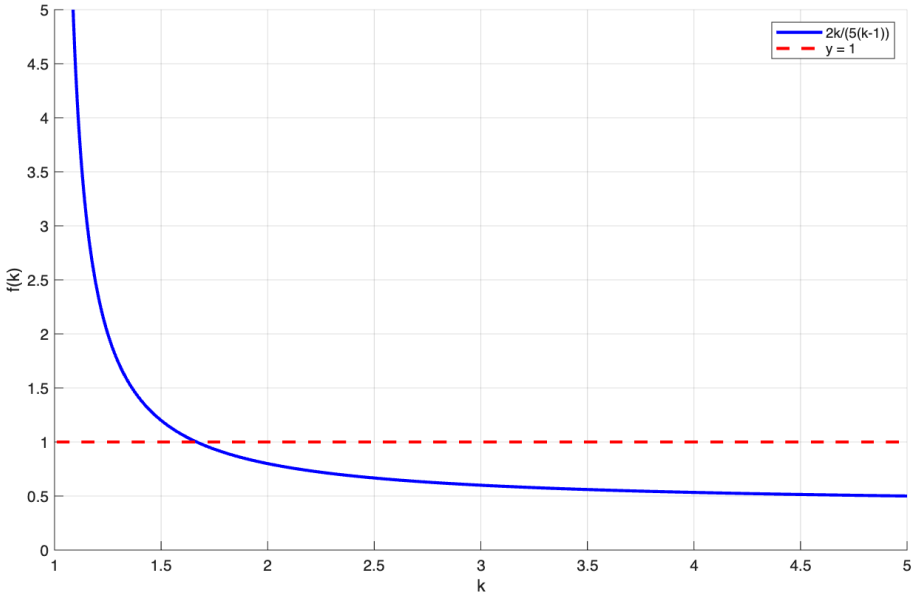


Figure 2. Plot of $\|\beta W_2(0)T(0)\| = \frac{2k}{5(k-1)}$ as a function of k

Finally, consider the Nyquist plot for different values of k.

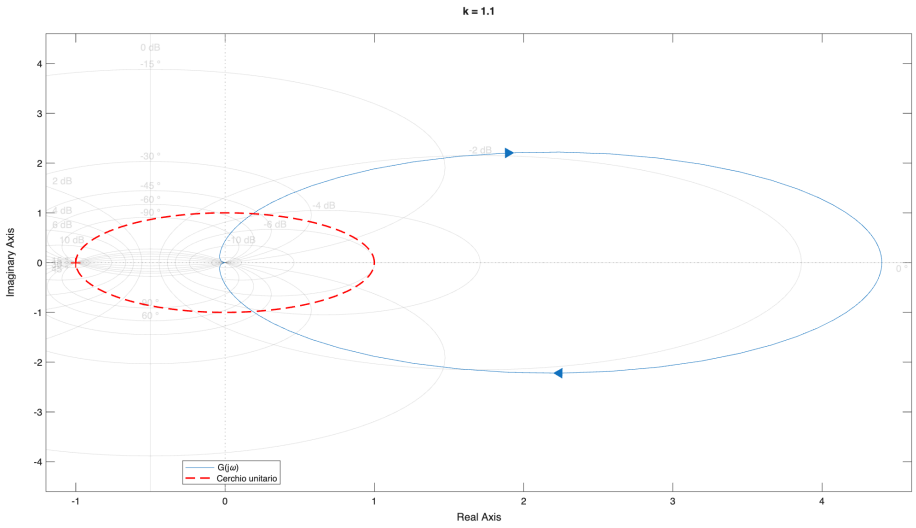


Figure 3. For $k = 1.1$, the Nyquist plot crosses the unit circle.

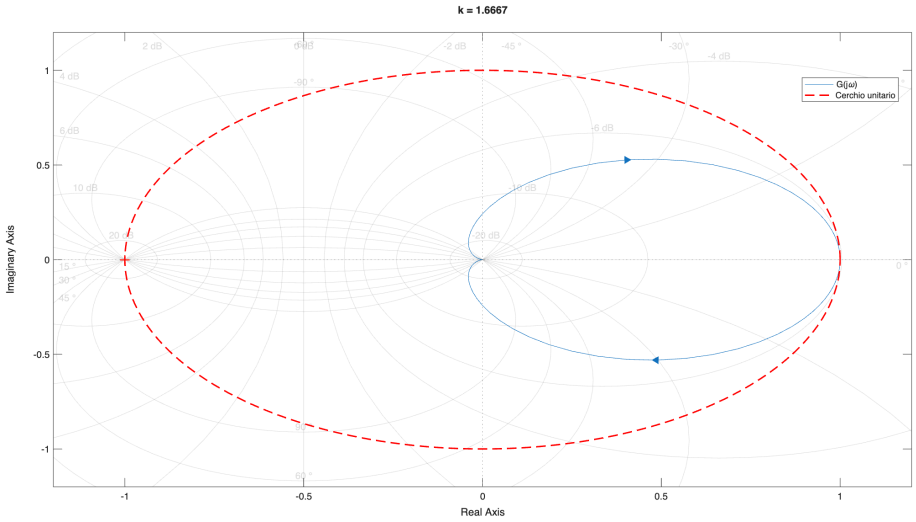


Figure 4. For $k = \frac{5}{3}$, the Nyquist plot is tangent to the unit circle.

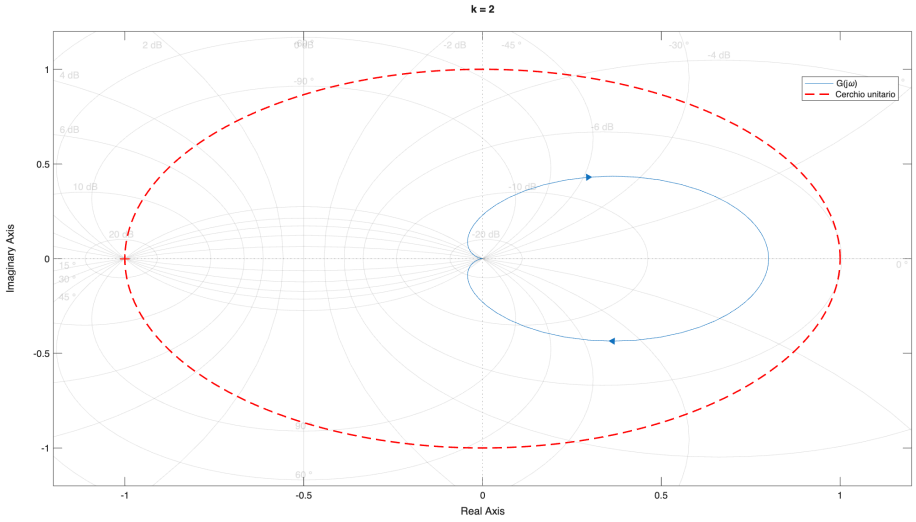


Figure 5. For $k = 2$, the Nyquist plot does not intersect the unit circle.

6 Performance

We now seek the value of k that minimises

$$\alpha_{\min} = \sup_{\omega} \|\bar{S}(j\omega)W_1(j\omega)\|$$

where

$$\begin{aligned}
\bar{S} &= \frac{1}{1+\bar{L}} \\
&= \frac{1}{1+L(1+\Delta W_2)} \\
&= \frac{1}{1+L+L\Delta W_2} \\
&= \frac{1}{(1+L)\Big(1+\frac{L}{1+L}\Delta W_2\Big)} \\
&= \frac{1}{1+L}\cdot\frac{1}{1+T\,\Delta\,W_2} \\
&= \frac{S}{1+T\,\Delta\,W_2}
\end{aligned}$$

We obtain

$$\|\bar{S}W_1\| = \left\|\frac{S}{1+T\,\Delta\,W_2}\cdot W_1\right\|$$

Since $\|\Delta\|_\infty \le 2$,

$$\begin{aligned}
1 &= \|1+T\,\Delta\,W_2 - T\,\Delta\,W_2\| \\
&\le \|1+T\,\Delta\,W_2\| + \|T\,\Delta\,W_2\| \\
&\le \|1+T\,\Delta\,W_2\| + 2\|TW_2\| \\
1-2\|TW_2\| &\le \|1+T\,\Delta\,W_2\| \\
\frac{1}{\|1+T\,\Delta\,W_2\|} &\le \frac{1}{1-2\|TW_2\|}
\end{aligned}$$

in the worst case,

$$\begin{aligned}
\|\bar{S}W_1\| &= \left\|\frac{S}{1+T\,\Delta\,W_2}\cdot W_1\right\| \\
&= \frac{\|S\cdot W_1\|}{\|1+T\,\Delta\,W_2\|} \\
&\le \frac{\|S\,W_1\|}{1-2\|TW_2\|}
\end{aligned}$$

where

$$\begin{aligned}
\|S(j\omega)W_1(j\omega)\| &= \left\|\frac{s-1}{s+k-1}\cdot\frac{1}{s+1}\right\|_{s=j\omega} \\
&= \left\|\frac{j\omega-1}{j\omega+k-1}\cdot\frac{1}{j\omega+1}\right\| \\
&= \sqrt{\frac{\omega^2 \not+ 1}{(\omega^2+(k-1)^2)(\omega^2 \not+ 1)}} \\
&= \sqrt{\frac{1}{\omega^2+(k-1)^2}}
\end{aligned}$$

and

$$\begin{aligned}
\|T(j\omega)W_2(j\omega)\| &= \left\|\frac{k}{s+k-1}\cdot\frac{2}{s+10}\right\|_{s=j\omega} \\
&= \left\|\frac{k}{j\omega+k-1}\cdot\frac{2}{j\omega+10}\right\| \\
&= \frac{2k}{\sqrt{(\omega^2+(k-1)^2)(\omega^2+100)}} \\
&\Downarrow \\
1-2\|T(j\omega)W_2(j\omega)\| &= 1-\frac{4k}{\sqrt{(\omega^2+(k-1)^2)(\omega^2+100)}} \\
&= \frac{\sqrt{(\omega^2+(k-1)^2)(\omega^2+100)}-4k}{\sqrt{(\omega^2+(k-1)^2)(\omega^2+100)}} \\
&\Downarrow \\
\frac{1}{1-2\|T(j\omega)W_2(j\omega)\|} &= \frac{\sqrt{(\omega^2+(k-1)^2)(\omega^2+100)}}{\sqrt{(\omega^2+(k-1)^2)(\omega^2+100)}-4k}
\end{aligned}$$

It follows that

$$\begin{aligned}
\frac{\|SW_1\|}{1-2\|TW_2\|} &= \sqrt{\frac{1}{\omega^2+(k-1)^2}}\,\frac{\sqrt{\omega^2+(k-1)^2}\sqrt{(\omega^2+100)}}{\sqrt{(\omega^2+(k-1)^2)(\omega^2+100)}-4k} \\
&= \frac{\sqrt{\omega^2+100}}{\sqrt{(\omega^2+(k-1)^2)(\omega^2+100)}-4k}
\end{aligned}$$

To find the value of ω, denoted by $\bar{\omega}$, at which $\frac{\|SW_1\|}{1-2\|TW_2\|}$ attains its maximum, differentiate with respect to ω. This gives

$$\begin{aligned}
\frac{d}{d\omega}\frac{\|SW_1\|}{1-2\|TW_2\|} &= \frac{d}{d\omega}\left(\frac{\sqrt{\omega^2+100}}{\sqrt{(\omega^2+(k-1)^2)(\omega^2+100)}-4k}\right) \\
&= \frac{\left(\frac{d}{d\omega}\sqrt{\omega^2+100}\right)\left(\sqrt{(\omega^2+(k-1)^2)(\omega^2+100)}-4k\right)-\sqrt{\omega^2+100}\,\frac{d}{d\omega}\left(\sqrt{(\omega^2+(k-1)^2)(\omega^2+100)}-4k\right)}{\left(\sqrt{(\omega^2+(k-1)^2)(\omega^2+100)}-4k\right)^2}
\end{aligned}$$

where

$$\begin{aligned}
\frac{d}{d\omega}\left(\sqrt{\omega^2+100}\right) &= \frac{\cancel{2}\,\omega}{\cancel{2}\sqrt{\omega^2+100}} \\
&= \frac{\omega}{\sqrt{\omega^2+100}}
\end{aligned}$$

and

$$\begin{aligned}
\frac{d}{d\omega}\left(\sqrt{(\omega^2+(k-1)^2)(\omega^2+100)}-4k\right) &= \frac{d}{d\omega}\left(\sqrt{\omega^4+100\omega^2+(k-1)^2\omega^2+100(k-1)^2}-4k\right) \\
&= \frac{1}{2}\frac{4\omega^3+200\omega+2(k-1)^2\omega}{\sqrt{\omega^4+100\omega^2+(k-1)^2\omega^2+100(k-1)^2}} \\
&= \frac{1}{\cancel{2}}\frac{\cancel{2}\,(2\omega^3+100\omega+(k-1)^2\omega)}{\sqrt{\omega^4+100\omega^2+(k-1)^2\omega^2+100(k-1)^2}} \\
&= \frac{2\omega^3+100\omega+(k-1)^2\omega}{\sqrt{\omega^4+100\omega^2+(k-1)^2\omega^2+100(k-1)^2}} \\
&= \frac{2\omega^3+100\omega+(k-1)^2\omega}{\sqrt{(\omega^2+(k-1)^2)(\omega^2+100)}}
\end{aligned}$$

It follows that the numerator is

$$\left(\frac{d}{d\omega}\sqrt{\omega^2+100}\right)\left(\sqrt{(\omega^2+(k-1)^2)(\omega^2+100)}-4k\right)-\sqrt{\omega^2+100}\,\frac{d}{d\omega}\left(\sqrt{(\omega^2+(k-1)^2)(\omega^2+100)}-4k\right) =$$

$$\frac{\omega}{\sqrt{\omega^2+100}}\left(\sqrt{(\omega^2+(k-1)^2)(\omega^2+100)}-4k\right)-\sqrt{\omega^2+100}\frac{2\omega^3+100\omega+(k-1)^2\omega}{\sqrt{(\omega^2+(k-1)^2)(\omega^2+100)}} =$$

$$\omega\left[\frac{1}{\sqrt{\omega^2+100}}\left(\sqrt{(\omega^2+(k-1)^2)(\omega^2+100)}-4k\right)-\sqrt{\omega^2+100}\frac{2\omega^2+100+(k-1)^2}{\sqrt{(\omega^2+(k-1)^2)(\omega^2+100)}}\right] =$$

$$\omega\frac{(\omega^2+(k-1)^2)(\omega^2+100)-4k\sqrt{(\omega^2+(k-1)^2)(\omega^2+100)}-(\omega^2+100)(2\omega^2+100+(k-1)^2)}{\sqrt{\omega^2+100}\sqrt{(\omega^2+(k-1)^2)(\omega^2+100)}}$$

Since the denominator is positive and $\omega>0$, we only need to examine the following term to determine whether the function is increasing or decreasing.

$$(\omega^2+(k-1)^2)(\omega^2+100)-4k\sqrt{(\omega^2+(k-1)^2)(\omega^2+100)}-(\omega^2+100)(2\omega^2+100+(k-1)^2)$$

Since $k>\frac{5}{3} \quad\Rightarrow\quad -k<-\frac{5}{3}$,

$$\begin{aligned}
&(\omega^2+(k-1)^2)(\omega^2+100)-4k\sqrt{(\omega^2+(k-1)^2)(\omega^2+100)}-(\omega^2+100)(2\omega^2+100+(k-1)^2) &<\\
&(\omega^2+(k-1)^2)(\omega^2+100)-4\frac{5}{3}\sqrt{(\omega^2+(k-1)^2)(\omega^2+100)}-(\omega^2+100)(2\omega^2+100+(k-1)^2) &=\\
&(\omega^2+100)\big(\omega^2+(k\not{-}1)^2-2\omega^2-100-(k\not{-}1)^2\big)-\frac{20}{3}\sqrt{(\omega^2+(k-1)^2)(\omega^2+100)} &=\\
&(\omega^2+100)(\omega^2-2\omega^2-100)-\frac{20}{3}\sqrt{(\omega^2+(k-1)^2)(\omega^2+100)} &=\\
&(\omega^2+100)(-\omega^2-100)-\frac{20}{3}\sqrt{(\omega^2+(k-1)^2)(\omega^2+100)} &=\\
&-(\omega^2+100)^2-\frac{20}{3}\sqrt{(\omega^2+(k-1)^2)(\omega^2+100)} &\leq\\
&0
\end{aligned}$$

for $k>\frac{5}{3}$, the function $\frac{\|SW_1\|}{1-2\|TW_2\|}$ decreases and therefore attains its maximum at $\omega=0$.

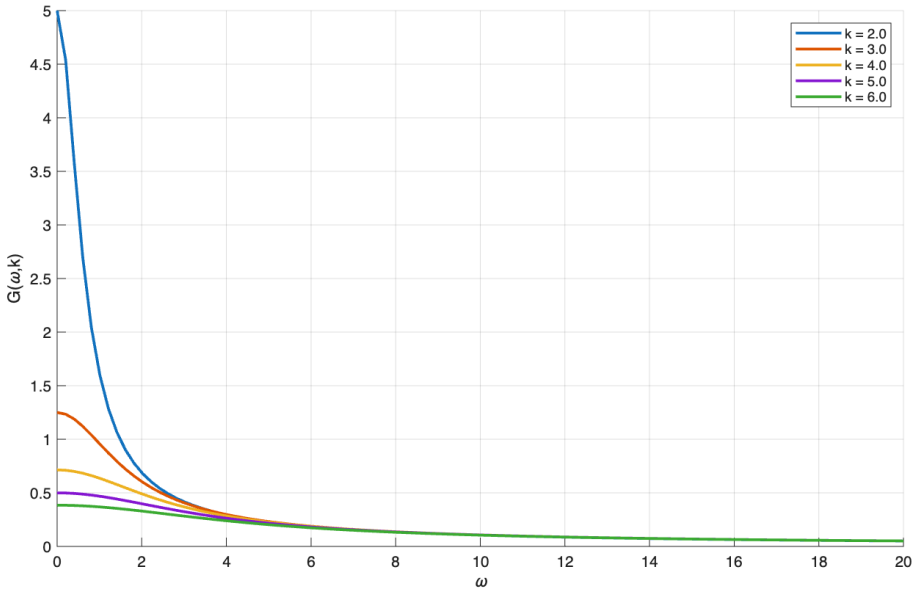


Figure 6. Plot of $\frac{\|SW_1\|}{1-2\|TW_2\|}$ for selected fixed values of k

It follows that

$$\begin{aligned}
\frac{\|SW_1\|_\infty}{1-2\|TW_2\|_\infty} &= \frac{\|S(0)W_1(0)\|}{1-2\|T(0)W_2(0)\|} \\
&= \left.\frac{\sqrt{\omega^2+100}}{\sqrt{\left(\omega^2+(k-1)^2\right)\left(\omega^2+100\right)}-4k}\right|_{\omega=0} \\
&= \frac{\sqrt{100}}{\sqrt{(k-1)^2 100}-4k} \\
&= \frac{10}{10(k-1)-4k} \\
&= \frac{10}{10k-10-4k} \\
&= \frac{10}{6k-10} \\
&= \frac{10}{2(3k-5)} \\
&= \frac{5}{3k-5}
\end{aligned}$$

Since

$$\begin{aligned}
\frac{d}{dk}\left(\frac{\|SW_1\|_\infty}{1-2\|TW_2\|_\infty}\right) &= \frac{d}{dk}\left(\frac{5}{3k-5}\right) \\
&= \frac{-5\cdot 3}{(3k-5)^2} \\
&= -\frac{15}{(3k-5)^2} \\
&< 0
\end{aligned}$$

as $k \to \infty$, $\|\bar{S}W_1\|$ approaches its infimum.

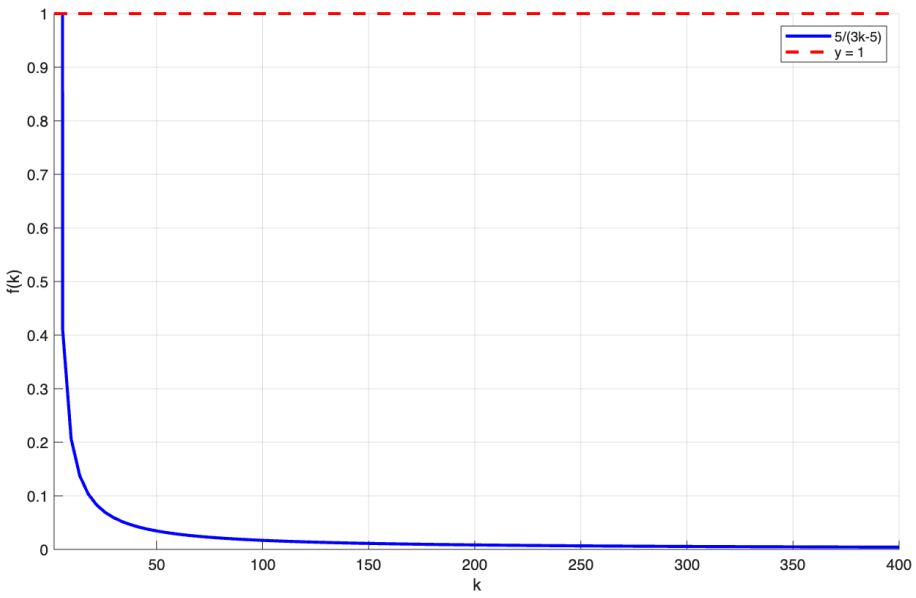


Figure 7. Plot of $\frac{\|S(0)\,W_1(0)\|}{1-2\|T\,(0)W_2(0)\|} = \frac{5}{3k-5}$ as a function of k